

class09

Maasfa Shaikh

```
candy_file <-  
"https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-ranking/candy-data.csv"  
  
candy = read.csv(candy_file, row.names=1)  
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard bar	pluribus	sugarpercent	pricepercent	winpercent	
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

```
#Q1. How many different candy types are in this dataset?  
dim(candy)
```

```
[1] 85 12
```

```
# there are 85 candy types in this dataset
```

```
#Q2. How many fruity candy types are in the dataset?
```

```
sum(candy$fruity)
```

```
[1] 38
```

```
# There are 38 fruity candy types
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
candy["Peanut butter M&M's", ]$winpercent
```

```
[1] 71.46505
```

my favorite candy is peanut butter m&ms and the winpercent value for that is:
71.46505

Q4. What is the winpercent value for “Kit Kat”?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

the winpercent for kitkats is 76.7686

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

the winpercent value for tootsie rolls is 49.6535

```
library("skimr")  
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	

numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

winpercent, pricepercent, and sugarpercent are on a continuous scale but the rest of the variables are binary

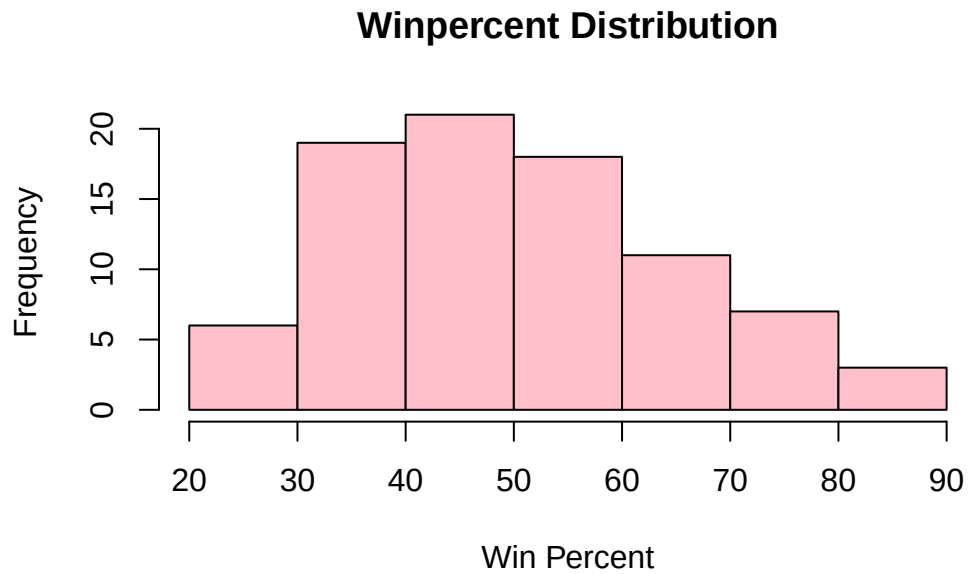
Q7. What do you think a zero and one represent for the candy\$chocolate column?

I think the zeros and ones represent yes or no / true or false for each question. For the candy\$chocolate column, the zeros would indicate that there is no chocolate, whereas the one would indicate that there is chocolate in the candy

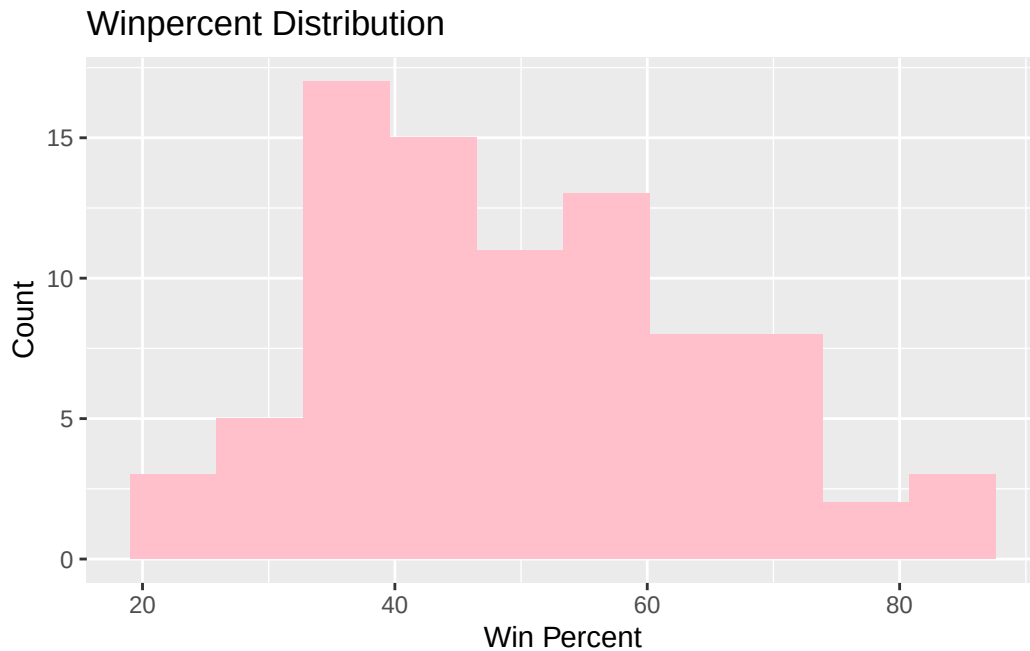
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
#base R
hist(candy$winpercent,
      col = "pink",
```

```
main = "Winpercent Distribution",  
xlab = "Win Percent")
```



```
#ggplot  
  
library(ggplot2)  
  
ggplot(candy, aes(x = winpercent)) +  
  geom_histogram(fill = "pink", bins = 10) +  
  labs(title = "Winpercent Distribution",  
        x = "Win Percent",  
        y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical? > The graphs are a little skewed.

Q10. Is the center of the distribution above or below 50%?

The center is below 50%

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

On average chocolate candies have a higher winpercent.

Q12. Is this difference statistically significant?

```
t.test(candy$winpercent[candy$chocolate == 1],
       candy$winpercent[candy$fruity == 1])
```

Welch Two Sample t-test

```
data: candy$winpercent[candy$chocolate == 1] and candy$winpercent[candy$fruity == 1]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

There is a statistically significant difference. The p-value is 2.871e-08, which is much lower than 0.5

Q13. What are the five least liked candy types in this set?

```
head(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard bar	pluribus	sugarpercent	pricepercent	
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The least liked candies are Nik L Nip, Boston baked beans, Chiclets, Super bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(-candy$winpercent), ], 5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisp	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0		0		0.720
Reese's Miniatures		0	0	0		0		0.034
Twix		1	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Snickers		0	0	1		0		0.546

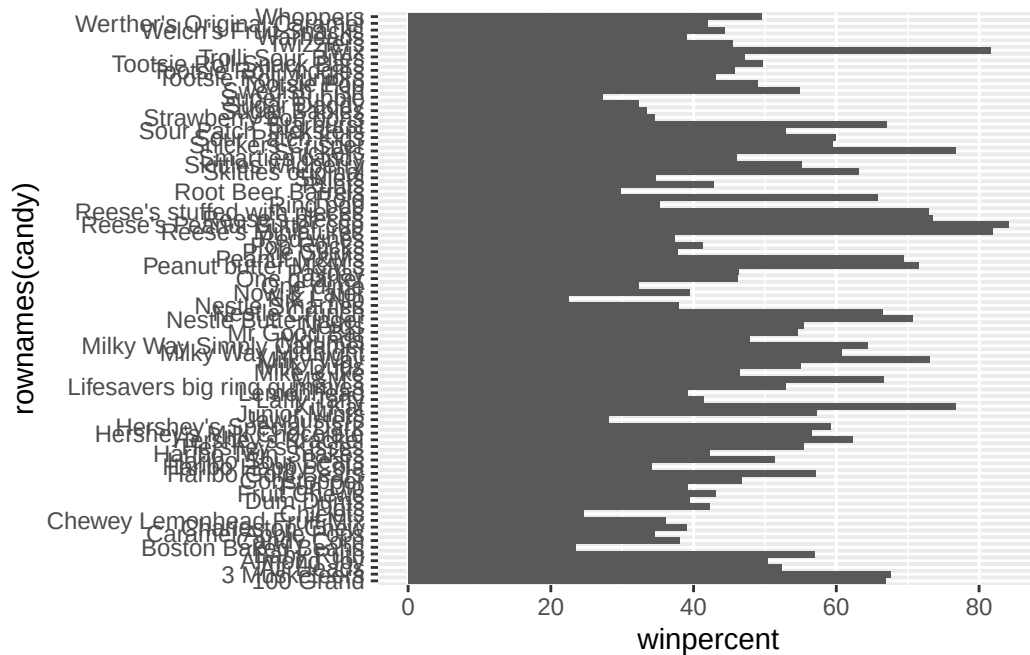
	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

The most liked candies are Reese's peanut butter cup, reese's miniatures, twix, kit kat, and snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

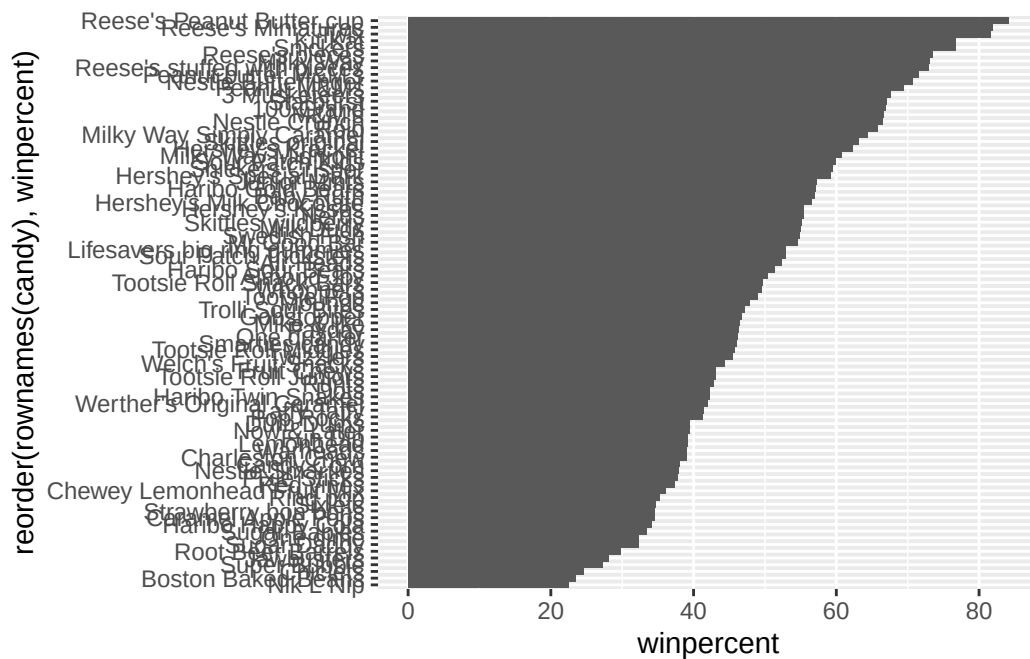
```
library(ggplot2)

ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



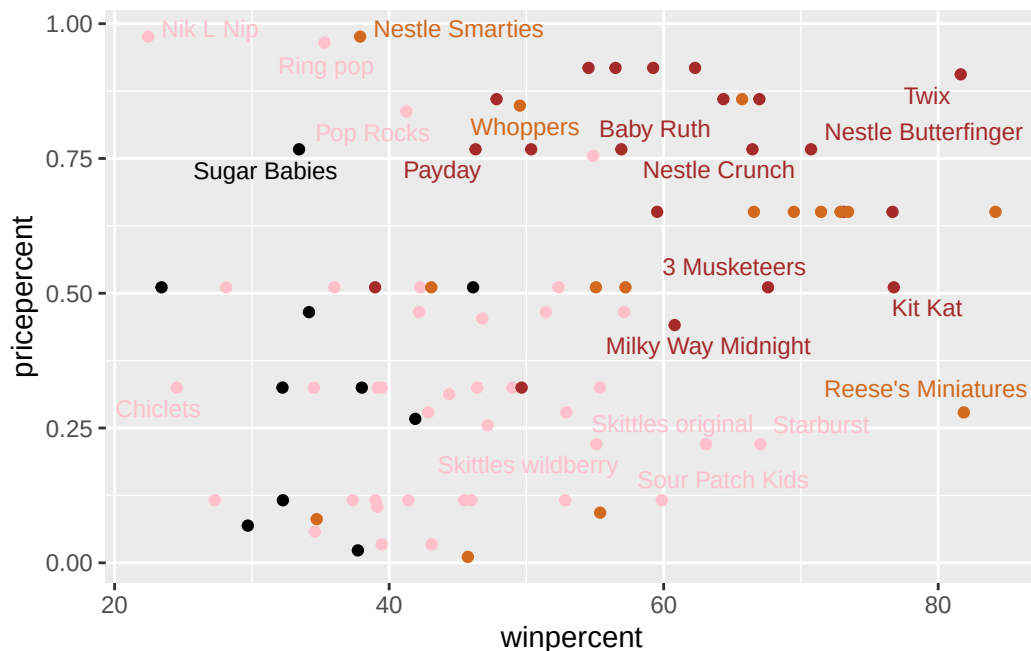
Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```

```
my_cols = rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[candy$bar == 1] = "brown"
my_cols[candy$fruity == 1] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```

Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck? > I think Reese's miniatures are the candy that are the least expensive and also have a high winpercent. It is chocolate.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

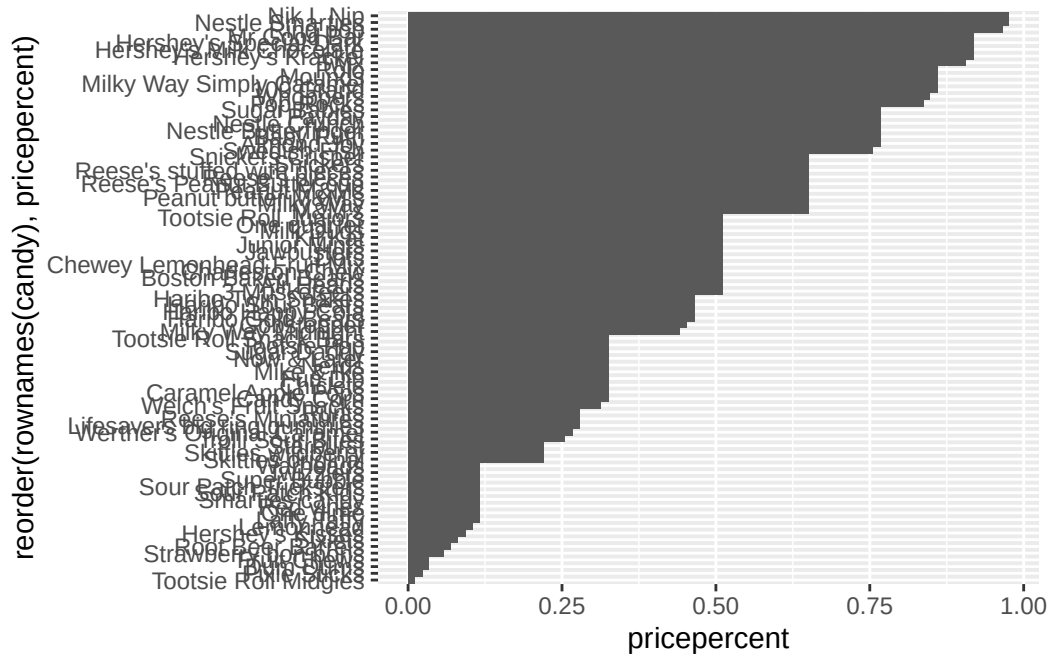
```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive candy types in the dataset is Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's milk chocolate. The least popular one out of these is Nik L Nip with a winpercent of 22.44534

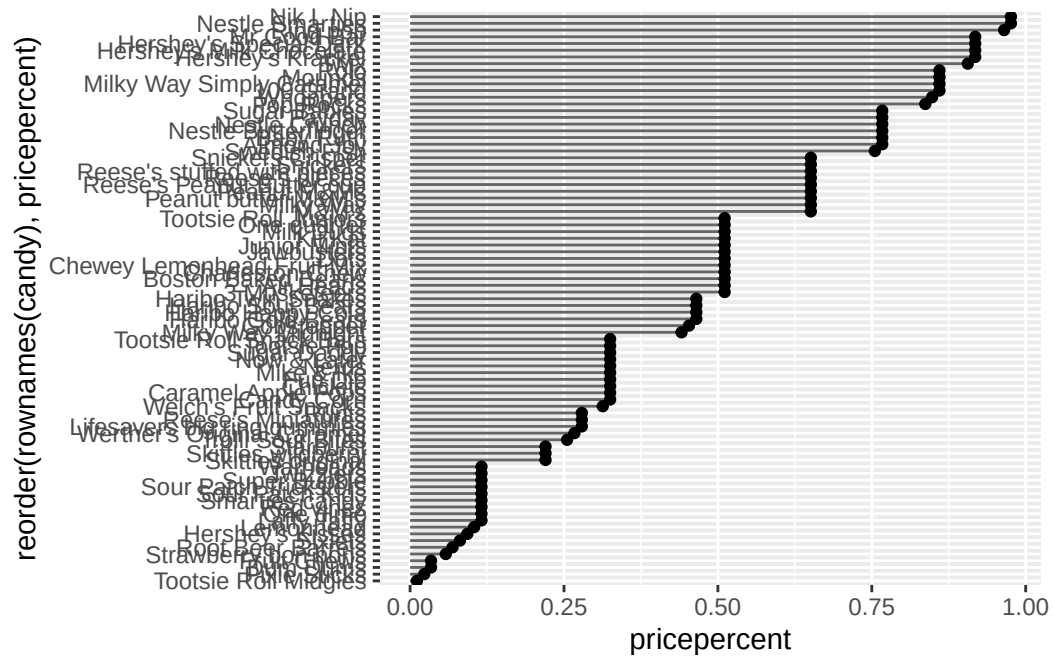
Q21. Make a barplot again with `geom_col()` this time using `pricepercent` and then improve this step by step, first ordering the x-axis by value and finally making a so called "dot chat" or "lollipop" chart by swapping `geom_col()` for `geom_point()` + `geom_segment()`.

```
ggplot(candy) +
  aes(x = reorder(rownames(candy), pricepercent),
      #reordered so its cleaner going from cheapest to most expensive
      y = pricepercent) +
  geom_col() +
  coord_flip()
```



```
#flipped it to the side
```

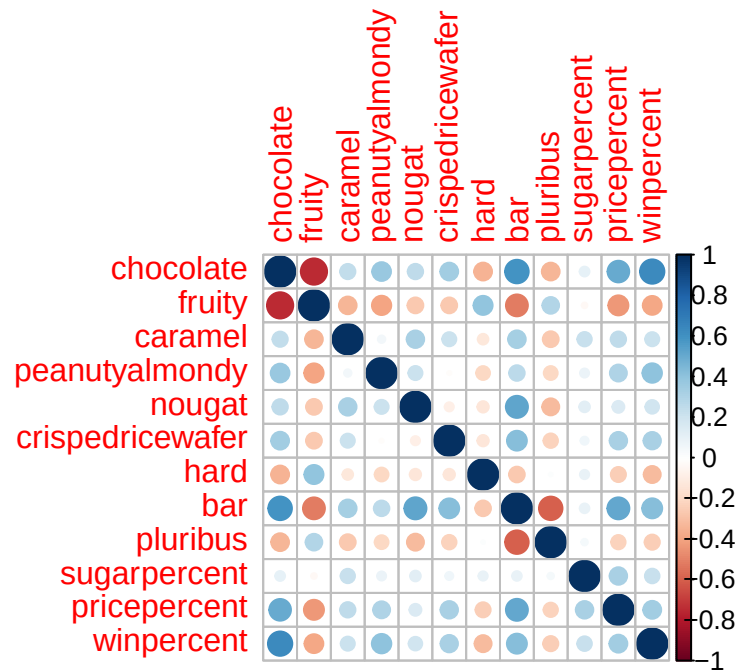
```
#making lollipop chart
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                        xend = 0), col="gray40") +
  geom_point()
```



```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)? > Chocolate and fruity candy variables are anti-correlated

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)? > I think the variables that would have the biggest contribution to PC1 would be chocolate, bar, and peanut almondy

```
pca <- prcomp(candy, scale = TRUE)
```

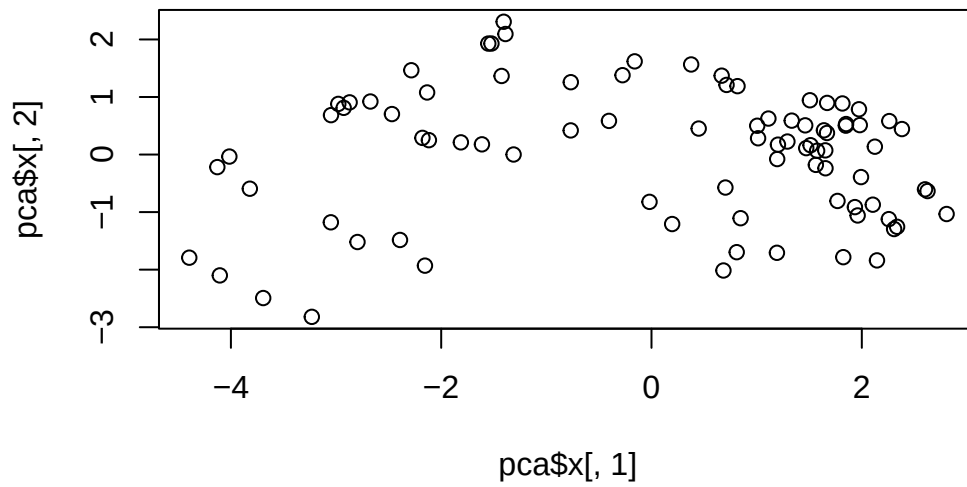
```
summary(pca)
```

Importance of components:

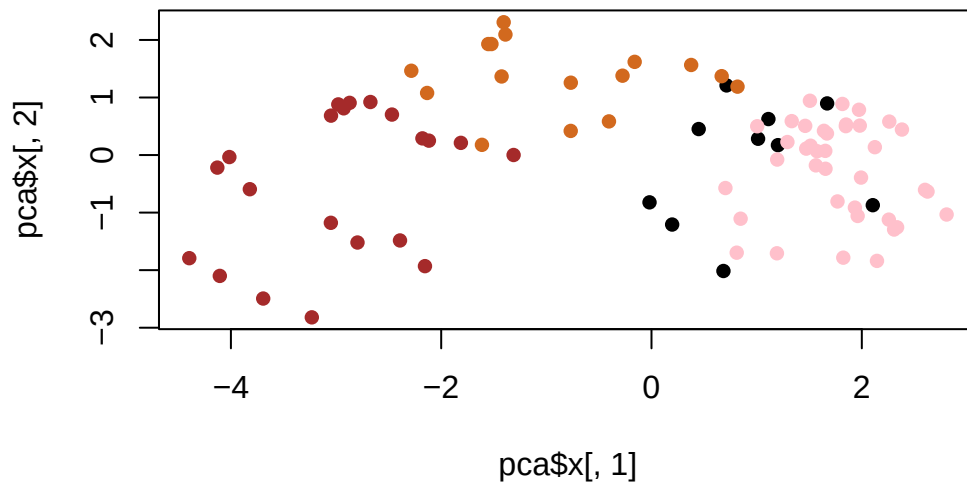
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1], pca$x[,2])
```



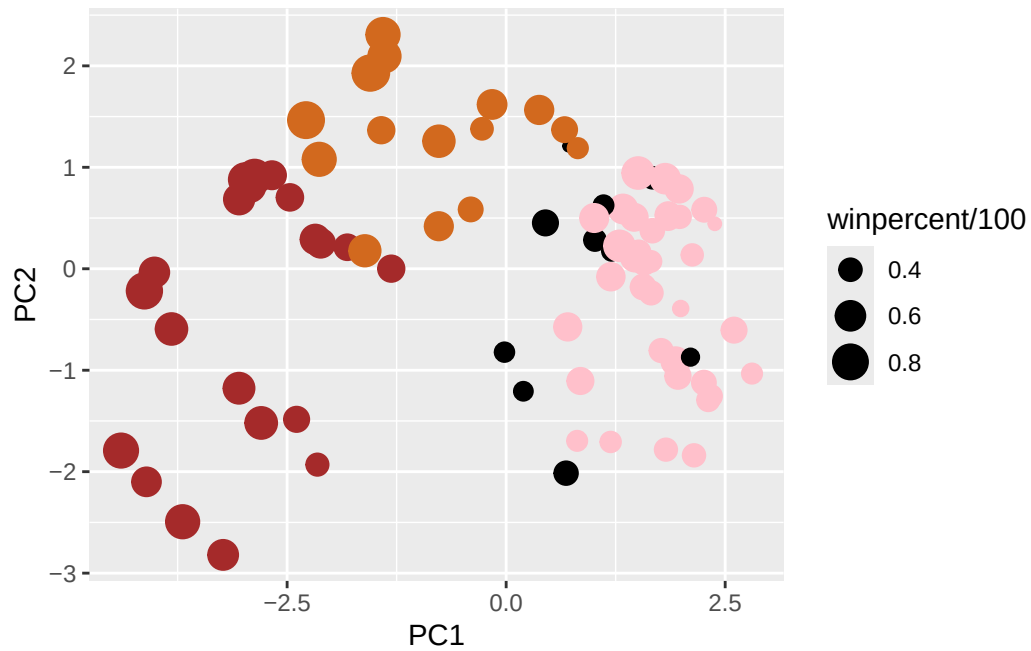
```
#adding character and color  
plot(pca$x[,1], pca$x[,2], col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)

p
```

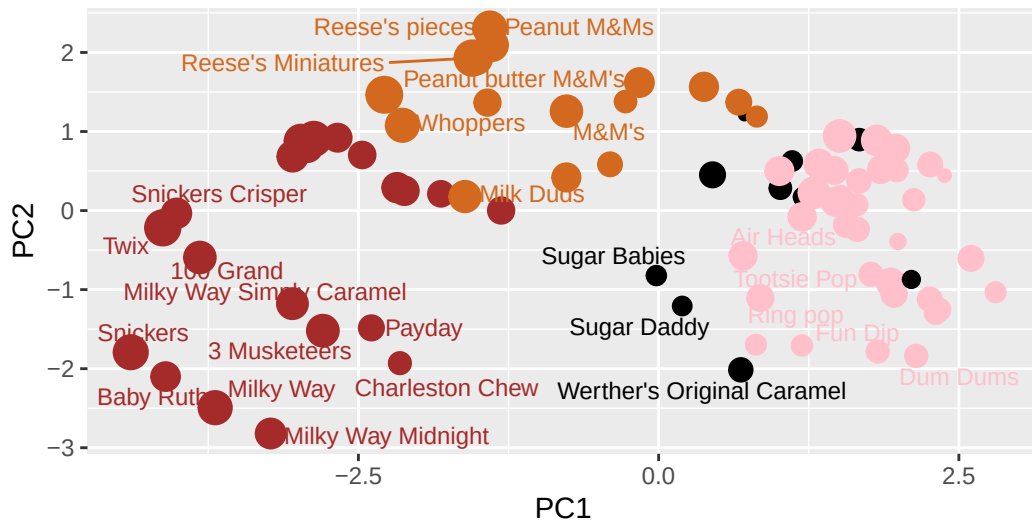



```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Halloween Candy PCA Space

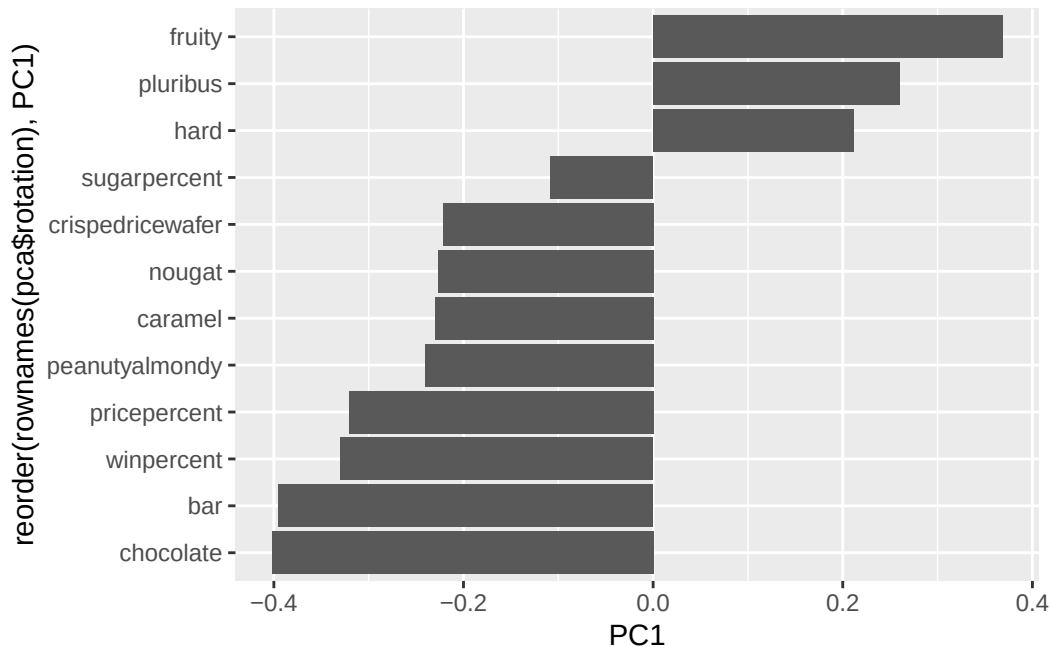
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(x = reorder(rownames(pca$rotation), PC1),
      y = PC1) +
  geom_col() +
  coord_flip()
```



the fruity and pluribus variables are picked up strongly by PC 1 in the positive direction. This makes sense because those variables were strongly associated with each other in the correlation graph. Fruity candies are usually hard and come in a bag with multiples (pluribus).

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

it seems that the winning candy has a combination of being a chocolate, bar, have peanuty almondy and not be fruity. We see the correlation graph showing strong correlation between these variables in different combinations, we also see a strong contribution of these elements to PC1. And lastly, we also saw that chocolate candies have a high winpercent compared to the rest of the candy types.

```
losers = candy[which(candy$winpercent < 50),]
winners = candy[which(candy$winpercent >= 50),]

mean(winners$pricepercent)
```

```
[1] 0.580359
```

```
mean(losers$pricepercent)
```

```
[1] 0.3743696
```

```
t.test(winners$pricepercent, losers$pricepercent)
```

Welch Two Sample t-test

```
data: winners$pricepercent and losers$pricepercent
t = 3.5653, df = 82.798, p-value = 0.0006068
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.09107157 0.32090727
sample estimates:
mean of x mean of y
0.5803590 0.3743696
```

Q26. Are popular candies more expensive? In other words: is price significantly different between “winners” and “losers”? List both average values and a P-value along with your answer.

winners have a higher price compared to losers. The average value of price percent for winners is 0.58 whereas for losers it is 0.37. comparing them using a welch two sample t test also showed a p value of 0.0006068, which is statistically significant.

Q27. Are candies with more sugar more likely to be popular? What is your interpretation of the means and P-value in this case?

```
mean(winners$sugarpercent)
```

```
[1] 0.5343077
```

```
mean(losers$sugarpercent)
```

```
[1] 0.4314565
```

```
t.test(winners$sugarpercent, losers$sugarpercent)
```

Welch Two Sample t-test

```
data: winners$sugarpercent and losers$sugarpercent
t = 1.6958, df = 81.799, p-value = 0.09373
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.01780701  0.22350935
sample estimates:
mean of x mean of y
0.5343077 0.4314565
```

candies with more sugar do not tend to be statistically significantly more popular compared to those with less sugar. The mean for popularity for winners of sugar percent was 0.5343, and the mean for popularity for losers of sugar percent was 0.4314. The p-value for the t test showed 0.09373, which is much bigger than 0.05, meaning there is no statistically significant difference in winners and losers of sugarpercent.